

# POC-07: Inventory Management & Procurement System

## Retail Domain

**Phase Weight Distribution:** P1=15% | P2=20% | P3=20% | P4=25% | P5=20% **Duration:** 25 working days | **Domain:** Retail Operations & Supply Chain

### 1. Business Context

India's retail sector manages millions of SKUs across organized and unorganized retail. Efficient inventory management — knowing what's on shelf, what's running low, and when to reorder — directly impacts revenue (stockouts lose sales) and working capital (overstock ties up cash). This POC builds the inventory management and procurement platform: tracking stock levels, raising purchase orders, managing supplier catalogs, and alerting on low-stock conditions.

### 2. Domain Personas

| Persona      | Role                | Uses the System To                                        |
|--------------|---------------------|-----------------------------------------------------------|
| Priya Sharma | Store Manager       | Monitor stock levels, approve purchase orders             |
| Raj Patel    | Inventory Analyst   | Analyze stock trends, set reorder points, audit movements |
| Anita Singh  | Procurement Officer | Raise and track purchase orders with suppliers            |
| Dev Kumar    | Warehouse Staff     | Record stock movements (in/out), update physical counts   |

### 3. Core Entities & Data Model

#### Product

| Field           | Type            | Description                                                  |
|-----------------|-----------------|--------------------------------------------------------------|
| id              | Integer (PK)    | Auto-generated                                               |
| sku             | String (unique) | e.g., SKU-GRO-0042                                           |
| name            | String          | Product name                                                 |
| category        | Enum            | grocery / electronics / clothing / household / personal_care |
| unit_price      | Float           | Selling price in ₹                                           |
| cost_price      | Float           | Purchase cost in ₹                                           |
| unit_of_measure | String          | pieces / kg / litre / box                                    |
| reorder_point   | Integer         | Stock level that triggers reorder alert                      |

| Field            | Type          | Description                       |
|------------------|---------------|-----------------------------------|
| reorder_quantity | Integer       | Quantity to order when reordering |
| supplier_id      | FK → Supplier | Primary supplier                  |

## StockLevel

| Field              | Type           | Description                 |
|--------------------|----------------|-----------------------------|
| id                 | Integer (PK)   | Auto-generated              |
| product_id         | FK → Product   | Product                     |
| warehouse_id       | FK → Warehouse | Location                    |
| quantity_on_hand   | Integer        | Current physical stock      |
| quantity_reserved  | Integer        | Reserved for pending orders |
| quantity_available | Integer        | on_hand - reserved          |
| last_updated       | DateTime       | Last stock update           |

## StockMovement

| Field            | Type              | Description                                     |
|------------------|-------------------|-------------------------------------------------|
| id               | Integer (PK)      | Auto-generated                                  |
| product_id       | FK → Product      | Product moved                                   |
| movement_type    | Enum              | receipt / sale / adjustment / transfer / return |
| quantity         | Integer           | Units moved (positive = in, negative = out)     |
| reference_number | String            | PO number or sale reference                     |
| notes            | String (nullable) | Reason for adjustment                           |
| recorded_at      | DateTime          | Timestamp                                       |
| recorded_by      | String            | Staff member                                    |

## PurchaseOrder

| Field        | Type            | Description                                             |
|--------------|-----------------|---------------------------------------------------------|
| id           | Integer (PK)    | Auto-generated                                          |
| po_number    | String (unique) | e.g., PO-2026-0042                                      |
| supplier_id  | FK → Supplier   | Supplier                                                |
| status       | Enum            | draft / submitted / acknowledged / received / cancelled |
| total_amount | Float           | Total order value in ₹                                  |

| Field             | Type                  | Description            |
|-------------------|-----------------------|------------------------|
| order_date        | Date                  | When PO was raised     |
| expected_delivery | Date                  | Expected delivery date |
| received_date     | Date (nullable)       | Actual receipt date    |
| items             | Relationship → POItem | Line items             |

## POItem

| Field             | Type               | Description          |
|-------------------|--------------------|----------------------|
| id                | Integer (PK)       | Auto-generated       |
| po_id             | FK → PurchaseOrder | Parent PO            |
| product_id        | FK → Product       | Product ordered      |
| quantity_ordered  | Integer            | Units requested      |
| unit_cost         | Float              | Agreed cost per unit |
| quantity_received | Integer (nullable) | Actually received    |

## Supplier

| Field              | Type            | Description                |
|--------------------|-----------------|----------------------------|
| id                 | Integer (PK)    | Auto-generated             |
| name               | String          | Supplier name              |
| supplier_code      | String (unique) | e.g., SUP-0001             |
| contact_email      | String          | Contact email              |
| payment_terms_days | Integer         | Net payment days           |
| lead_time_days     | Integer         | Typical delivery lead time |
| is_active          | Boolean         | Active supplier flag       |

## Alert (Stock)

| Field       | Type         | Description                                              |
|-------------|--------------|----------------------------------------------------------|
| id          | Integer (PK) | Auto-generated                                           |
| product_id  | FK → Product | Product                                                  |
| alert_type  | Enum         | low_stock / out_of_stock / overstock / reorder_suggested |
| message     | String       | Alert description                                        |
| is_resolved | Boolean      | Alert acknowledged                                       |

| Field        | Type     | Description         |
|--------------|----------|---------------------|
| triggered_at | DateTime | Alert creation time |

## 4. Business Rules

1. **Low stock alert:**  $\text{quantity\_available} \leq \text{reorder\_point} \rightarrow \text{create low\_stock alert}$
2. **Out of stock:**  $\text{quantity\_available} = 0 \rightarrow \text{create out\_of\_stock alert (critical)}$
3. **PO number format:** PO-{YEAR}-{NNNN} e.g., PO-2026-0042
4. **SKU format:** SKU-{CATEGORY\_PREFIX}-{NNNN} e.g., SKU-GRO-0042
5. **Stock movement:** quantity\_on\_hand updated after every StockMovement record
6. **Receiving PO:** PATCH /orders/{id}/receive triggers StockMovement(receipt) for each POItem

## 5. REST API Endpoints

| Method | Endpoint                       | Description                                        |
|--------|--------------------------------|----------------------------------------------------|
| GET    | /api/v1/products               | List products (filter by category, low_stock=true) |
| POST   | /api/v1/products               | Create new product                                 |
| GET    | /api/v1/products/{id}          | Get product with stock level                       |
| PATCH  | /api/v1/products/{id}/stock    | Update stock quantity (record movement)            |
| POST   | /api/v1/orders                 | Create purchase order                              |
| GET    | /api/v1/orders                 | List orders (filter by status, supplier)           |
| GET    | /api/v1/orders/{id}            | Get PO with line items                             |
| PATCH  | /api/v1/orders/{id}/receive    | Mark PO received, update stock                     |
| GET    | /api/v1/stock/low-alerts       | Get all low stock and out-of-stock products        |
| GET    | /api/v1/suppliers/{id}/catalog | Get supplier's products with cost prices           |
| GET    | /api/v1/dashboard              | Inventory dashboard                                |
| POST   | /api/v1/auth/register          | Register user                                      |
| POST   | /api/v1/auth/login             | Login                                              |

## 6. Phase 5 Agents

**Agents:** Demand Forecaster → Reorder Agent → Supplier Coordinator → Inventory Auditor

**State:** `InventoryAnalysisState` with fields: product\_id, product\_data, demand\_forecast, reorder\_recommendation, supplier\_quote, audit\_report, analysis\_status, errors, messages

## 7. Glossary

| Term            | Definition                                                  |
|-----------------|-------------------------------------------------------------|
| SKU             | Stock Keeping Unit — unique product identifier              |
| Reorder Point   | Stock level at which replenishment should begin             |
| Lead Time       | Days from PO submission to stock receipt                    |
| Stockout        | Zero inventory — unable to fulfill demand                   |
| Overstock       | Excess inventory — capital tied up unnecessarily            |
| FIFO            | First In, First Out — inventory accounting method           |
| Safety Stock    | Buffer inventory held to prevent stockouts                  |
| PO              | Purchase Order — formal request to supplier to supply goods |
| Stock Movement  | Any change in inventory quantity                            |
| Capacity Factor | Ratio of actual output to maximum possible output           |